Mohammad Uzair Shah

Email: mshah17@tennessee.edu | Cell: (865) 398-0062

Portfolio | LinkedIn

EDUCATION

Ph.D., Energy Science and Engineering Fall 2022 – Expected Fall 2026
Bredesen Center, University of Tennessee, Knoxville, TN, USA.
CGPA: 3.97/4.00, Percentage: 99 %

Master of Science in Energy Systems Engineering Fall 2020 – Summer 2022
National University of Science and Technology (NUST), Islamabad, Pakistan
CGPA: 3.85/4.00, Percentage: 98 %

Bachelor of Science in Mechanical Engineering Fall 2015 – March 2020
Pakistan Institute of Engineering and Applied Sciences (PIEAS), Islamabad, Pakistan
CGPA: 3.56/4.00, Percentage: 89 %

Exchange School (U.S. Department of State sponsored), Mechanical Engineering Fall 2018
University of Missouri, Columbia, MO, USA
CGPA: 3.63/4.00, Percentage: 91 %

WORK EXPERIENCE

Graduate Research Assistant, Biomass Conversion and Modeling Lab, UTK Fall 2022 - present

- Developed a Python-based, GIS-integrated decision support framework to optimize Mxg feedstock supply and site selection for SAF biorefineries.
- Performed spatial clustering and feedstock aggregation analysis to define viable supply radii for commercial-scale SAF facilities.
- Designed a system-level modeling platform integrating TEA and LCA to evaluate cost, emissions, and scalability of SAF production pathways.
- Applied multi-criteria spatial optimization using land use, soil, climate, supply chain and logistics data to identify high-potential SAF feedstock regions and minimize supply chain costs.
- Quantified regional economic impacts of SAF deployment using Input-Output modeling, including job creation and local value generation.
- Incorporated socio-economic and energy justice indicators to prioritize equitable and community-aligned SAF infrastructure development.

Research Projects

Data-Driven Integrated Decision Support Modeling Framework for Bioeconomy Opportunity Zones Analysis, Dissertation project

- Developing a Python-based, GIS-enabled decision support framework integrating TEA, LCA, and socio-economic metrics to optimize SAF feedstock production and biorefinery siting.
- Applying multi-criteria spatial analysis to balance cost, emissions, land suitability, and logistics for scalable and sustainable SAF supply chains.
- Incorporating social feasibility and energy justice indicators to ensure equitable deployment of SAF infrastructure and maximize community benefits.
- Translating model outcomes into actionable insights to inform SAF-related policy and support aviation decarbonization strategies.

Reducing Agricultural Carbon Intensity and Protecting Algal Crops (RACIPAC), DOE funded project

- Investigated feedstock–process–biochar–soil interactions to enhance carbon sequestration and reduce lifecycle emissions in bioenergy systems supporting SAF production across the Southeast.
- Evaluated the impact of biochar and poultry litter amendments on Mxg and biomass sorghum as low-carbon SAF feedstocks using machine learning, TEA, and LCA.
- Quantified carbon intensity reduction pathways and agronomic performance to support scalable, sustainable SAF supply chains.

A techno-economic and socio-environmental planning of wind turbines for sustainable development in Pakistan, Master's thesis project

- Integrated traditional techno-economic and life-cycle analysis with the crucial quantitative and qualitative social factors for sustainable siting of wind farms while recommending policy suggestions to improve development and social acceptance of wind farms in the community.

PUBLICATIONS

Peer-reviewed Journal Articles

- Shah, M. U., Khanum, S., Waqas, A., Janjua, A. K., & Shakir, S. (2023). A techno-economic and socio-environmental planning of wind farms for sustainable development and transition to a decarbonized scenario: Pakistan as a case study. *Sustainable Energy Technologies and Assessments*, *55*, 102969, https://doi.org/10.1016/j.seta.2022.102969.

CONFERENCE PRESENTATIONS

- **"Integrated GIS-based decision support model for optimal biomass preprocessing site identification using sociodemographic, economic, and environmental analysis"** (Oral presentation)
  American Society of Agricultural and Biological Engineers (ASABE) Annual International Meeting, Anaheim, CA, July 2024
- "**Identification of potential Miscanthus x giganteus (Mxg) production areas in Tennessee using an integrated GIS-based multicriteria decision support system modeling**" (Poster presentation)
  S1075 Multi-state Project 2024 Annual Meeting, Brookings, SD, July 2024

- "**GIS-based multi-criteria decision support for energy crop system analysis**" (Poster presentation)
  Frontiers in Biorefining, St. Simons Island, GA, September 2024

LEADERSHIP EXPERIENCE

Cultural ambassador, U.S. Department of State, Washington, DC — Fall 2018

- Participated in cultural exchange program UGRAD (1% acceptance rate) sponsored by U.S. Department of State to portray positive image of both countries and share cultural aspects.

President, PIEAS Thematic Society (PTS) — 2017-2018

- Headed various inter- and intra-university events at PIEAS including Synergy'17, PION'18 and Megafest'17 for community outreach and development.

SKILLS

Python, ArcGIS (geospatial analysis software), data curation and analysis, process modeling, renewable energy siting, socio-economic analysis, I-O modeling, software development, techno-economic analysis, life cycle assessment, survey design, community outreach, qualitative interviewing